

MECH5200 Numerical Methods for PDEs : *Video 24: One Dimensional FEM Boundary Conditions and Two Dimensional FEM*

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1D FEA Review

- The equation for the weak form for 1-D FEA is:

$$\left[w \frac{\partial u}{\partial x} \right]_{x_L}^{x_R} - \int_{x_L}^{x_R} \frac{\partial w}{\partial x} \cdot \frac{\partial u}{\partial x} dx = \int_{x_L}^{x_R} (wf) dx \quad (1)$$

- The discrete form of this equation for the weighting function associated with node i is:

$$\left[\phi_i \frac{\partial u_j}{\partial x} \right]_{x_L}^{x_R} - \sum_j u_j \int_{x_L}^{x_R} \frac{\partial \phi_i}{\partial x} \cdot \frac{\partial \phi_j}{\partial x} dx = \int_{x_L}^{x_R} (\phi_i f) dx \quad (2)$$

- Part I Today: Boundary conditions for 1-D FEA problems

1D FEA BC's

- Three boundary conditions can exist:

① Dirichlet :

$$u(x_L) = L$$

$$u(x_R) = R$$

② Neumann :

$$\frac{\partial u}{\partial x}(x_L) = g_L$$

$$\frac{\partial u}{\partial x}(x_R) = g_R$$

③ Mixed : Combination of Dirichlet and Neumann

Dirichlet BC's

- Dirichlet BC's are also relatively simple to implement.
- Approach is the same as in Finite Differences

$$u_i = n \quad (3)$$

- Force the nodal value by setting matrix entries to zero and diagonal to one.
- Let's see this in MATLAB

Neumann BC's

- Consider the weak form of the equation:

$$\int_{x_L}^{x_R} (wf) dx = \left[w \frac{\partial u}{\partial x} \right]_{x_L}^{x_R} - \int_{x_L}^{x_R} \frac{\partial w}{\partial x} \cdot \frac{\partial u}{\partial x} dx \quad (4)$$

- The equation has a term that includes the solution derivative at the boundary: $\left[w \frac{\partial u}{\partial x} \right]_{x_L}^{x_R}$
- This asks us to multiply the partial derivative at the boundary with the basis function value at the boundary.

Neumann BC's

- We know the value of the weighting basis, w , at the boundary is $= 1$ when $i = \text{boundary node}$.
- We also know the value of the Neumann boundary condition:

$$\begin{aligned}\frac{\partial u}{\partial x}(x_L) &= g_L \\ \frac{\partial u}{\partial x}(x_R) &= g_R\end{aligned}$$

- Inserting the known Neumann boundary condition for the boundary nodes in the weak form equation, we get:

$$\left[w \frac{\partial u}{\partial x} \right]_{x_L}^{x_R} = [1 \cdot g]_{x_L}^{x_R} = g_R - g_L. \quad (5)$$

Neumann BC's

- When we implement this, we simply append (add/subtract) the value g in the correct RHS column location and the boundary condition is **naturally** satisfied (ie. we simply add this known value).
- I.e. we did not need to perform any modifications to the A-matrix to implement the Neumann boundary condition. For this problem, the **natural** boundary condition is the Neumann boundary condition (this is not always the case, equation dependent).
- Let's see this in MATLAB!

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1 Review 1D FEM Equations

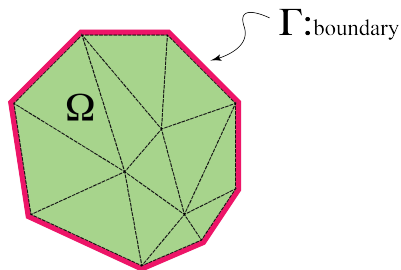
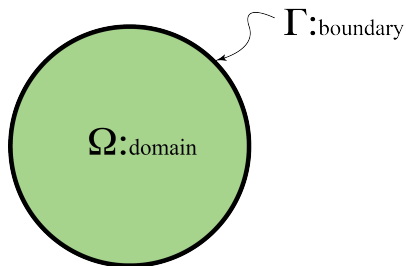
2 2D FEM

3 2D FEM

4 2-D Discretization

Definition of the domain

- The domain is Ω and the boundary of that domain is Γ :



Deriving the Weak Form in multiple dimensions

- The strong form of the equation on the domain Ω is:

$$\nabla^2 u = f \quad (6)$$

- With boundary conditions on the boundary Γ :

$$u(\Gamma_D) = h \quad (7)$$

$$\frac{\partial u}{\partial n}|_{\Gamma_N} = g \quad (8)$$

$$(9)$$

- We will multiply this equation by a weighting function w and then perform an integration over the domain:

$$\int_{\Omega} w \nabla^2 u d\Omega - \int_{\Omega} w f d\Omega = 0 \quad (10)$$

Deriving the Weak Form in multiple dimensions

- From previous slide:

$$\int_{\Omega} w \nabla^2 u d\Omega - \int_{\Omega} w f d\Omega = 0 \quad (11)$$

- We will use multi-dimensional integration by parts or *Green's theorem* to help reduce this to a useful weak form for FEM discretization:

$$\int_{\Omega} [\phi \nabla^2 \psi + \nabla \phi \cdot \nabla \psi] d\Omega = \int_{\Gamma} \phi (\nabla \psi \cdot \hat{n}) d\Gamma \quad (12)$$

- If we take $\phi = w$ and $\psi = u$, then:

$$\int_{\Omega} [w \nabla^2 u + \nabla w \cdot \nabla u] d\Omega = \int_{\Gamma} w (\nabla u \cdot \hat{n}) d\Gamma \quad (13)$$

2-D FEA

- If we insert equation 11 into equation 13, we get:

$$\int_{\Omega} \left[\underbrace{wf}_{w \nabla^2 u} + \nabla w \cdot \nabla u \right] d\Omega = \int_{\Gamma} w (\nabla u \cdot \hat{n}) d\Gamma \quad (14)$$

- The result is an equation that looks very similar to the 1-D derivation:

$$\int_{\Omega} [\nabla w \cdot \nabla u] d\Omega - \int_{\Gamma} w (\nabla u \cdot \hat{n}) d\Gamma = \int_{\Omega} -w f d\Omega \quad (15)$$

- Now all that is left is to discretize the problem and numerically solve the weak form above.

Discrete Equation

- Introduce a set of solution basis functions ϕ_j and a set of weighting or test functions ϕ_i .
- Setup an equation for each weighting function i :

$$\underbrace{\int_{\Omega} \left[\nabla \phi_i \cdot \nabla \sum_j u_j \phi_j \right] d\Omega}_{A \text{ Matrix}} - \underbrace{\int_{\Gamma} \phi_i (\nabla u_j \phi_j \cdot \hat{n}) d\Gamma}_{\text{natural B.C.'s (Neumann here)}} = \underbrace{\int_{\Omega} -\phi_i f d\Omega}_{R.H.S}$$

Discrete Equation

- Our resulting A -matrix will be composed of:

$$A_{i,j} = \int_{\Omega} [\nabla \phi_i \cdot \nabla \phi_j] d\Omega = ?? \quad (16)$$

- Our resulting RHS -vector will be composed of:

$$\int_{\Omega} -\phi_i f d\Omega = ?? \quad (17)$$

- Our resulting boundary conditions vector will be composed of:

$$\int_{\Gamma} \phi_i (\nabla u_j \phi_j \cdot \hat{n}) d\Gamma = ?? \quad (18)$$

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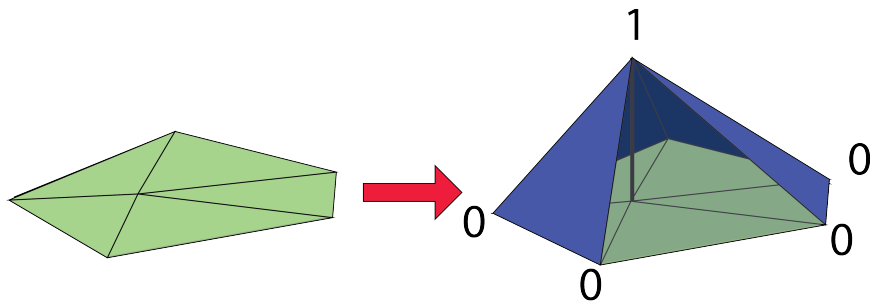
4 2-D Discretization

Discretizing the Domain

- We will use a triangulation T which contains both nodes and connectivity matrices.
- The FEA approach we will use will be to construct elemental matrices and stamp them in the global matrix (much like in 1-D).
- We will use linear hat basis functions

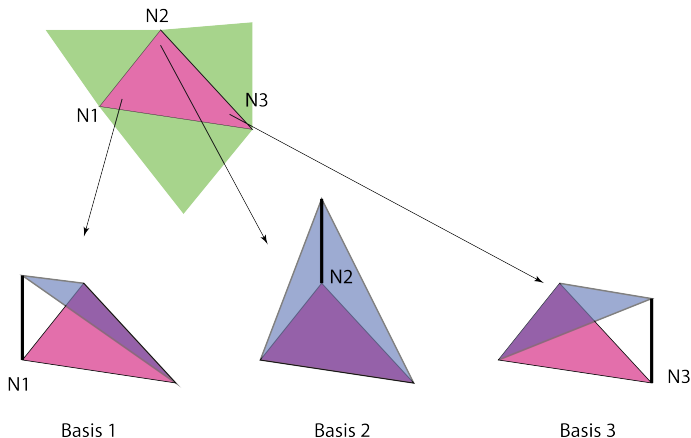
Basis Functions

- The nodal-hat basis function has a value of 1 at node i and a value of 0 elsewhere:



Basis Functions

- Each triangular element be associated with 3 nodal basis functions, one for each vertex of the triangle:
- The three basis functions for a triangle are:



Basis Functions

- The basis function ϕ_i can be represented as a plane

$$\phi_i = ax + by + c \quad (19)$$

- Where the a , b and c values are constants determined to satisfy the equation of the plane.
- The value plane has a value 1 at the node in question and zero at the other two nodes.

Basis Functions

- As such, the equations for the FIRST basis function on a triangle are:

$$\phi_{i=N1@N1} = 1 = a_{N1} \cdot x1 + b_{N1} \cdot y1 + c_{N1}$$

$$\phi_{i=N1@N2} = 0 = a_{N1} \cdot x2 + b_{N1} \cdot y2 + c_{N1}$$

$$\phi_{i=N1@N3} = 0 = a_{N1} \cdot x3 + b_{N1} \cdot y3 + c_{N1}$$

- Similarly, node 2 can be written as:

$$\phi_{i=N2@N1} = 0 = a_{N2} \cdot x1 + b_{N2} \cdot y1 + c_{N2}$$

$$\phi_{i=N2@N2} = 1 = a_{N2} \cdot x2 + b_{N2} \cdot y2 + c_{N2}$$

$$\phi_{i=N2@N3} = 0 = a_{N2} \cdot x3 + b_{N2} \cdot y3 + c_{N2}$$

Basis Functions

- The result is a linear system of equations that must be solved to get the coefficients of the plane for each basis function:

$$\begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix} \cdot \begin{pmatrix} a_{N1} \\ b_{N1} \\ c_{N1} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad (20)$$

- Similarly, for node 2:

$$\begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix} \cdot \begin{pmatrix} a_{N2} \\ b_{N2} \\ c_{N2} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad (21)$$

- Similarly, for node 3:

$$\begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix} \cdot \begin{pmatrix} a_{N3} \\ b_{N3} \\ c_{N3} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (22)$$

Basis Functions

- We can see that, for each basis function, there is a linear system that is solved for the basis function at each node i .
- The gradient of the basis function $\nabla\phi_i$ can be calculated as:

$$\nabla\phi_i = \left[\frac{\partial\phi_i}{\partial x}\hat{x}, \frac{\partial\phi_i}{\partial y}\hat{y} \right] = [a_i\hat{x}, b_i\hat{y}] \quad (23)$$

- Recalling that the A-Matrix is comprised of interactions between weighting and solution basis functions:

$$A_{i,j} = \int_{\Omega} [\nabla\phi_i \cdot \nabla\phi_j] d\Omega = ?? \quad (24)$$

- The integrand dot product gradient calculation is:

$$\nabla\phi_i \cdot \nabla\phi_j = [a_i\hat{i}, b_i\hat{j}] \cdot [a_j\hat{i}, b_j\hat{j}] = (a_i \cdot a_j) + (b_i \cdot b_j) \quad (25)$$

A Matrix Construction

This computation is done to interact the Nodal Solution Basis functions $N1$, $N2$ and $N3$ for a given triangle with the weighting basis functions for that same triangle:

$$A_{i,j} = \int_{\Omega} [\nabla \phi_i \cdot \nabla \phi_j] d\Omega = TriArea \cdot [a_i \hat{i}, b_i \hat{j}] \cdot [a_j \hat{i}, b_j \hat{j}] = (a_i \cdot a_j) + (b_i \cdot b_j) \quad (26)$$

$$A_{Elemental} = TriArea \cdot \begin{pmatrix} (a_{N1} \cdot a_{N1}) + (b_{N1} \cdot b_{N1}) & (a_{N1} \cdot a_{N2}) + (b_{N1} \cdot b_{N2}) & (a_{N1} \cdot a_{N3}) + (b_{N1} \cdot b_{N3}) \\ (a_{N2} \cdot a_{N1}) + (b_{N2} \cdot b_{N1}) & (a_{N2} \cdot a_{N2}) + (b_{N2} \cdot b_{N2}) & (a_{N2} \cdot a_{N3}) + (b_{N2} \cdot b_{N3}) \\ (a_{N3} \cdot a_{N1}) + (b_{N3} \cdot b_{N1}) & (a_{N3} \cdot a_{N2}) + (b_{N3} \cdot b_{N2}) & (a_{N3} \cdot a_{N3}) + (b_{N3} \cdot b_{N3}) \end{pmatrix} \quad (27)$$

These entries for each element can be stamped appropriately in the overall A-Matrix.

Right Hand Side

- Similarly, to construct the RHS:

$$RHS_i = \int_{\Omega} -\phi_i f d\Omega = ?? \quad (28)$$

- We can simplify for today – if $f = \text{constant}$, then this integral is:

$$RHS_i = \frac{-TriArea \cdot f}{3} \quad (29)$$

- Since the value of ϕ_i varies across the triangle.

Right Hand Side

- In class, we'll give this a try
- Solve:

$$\nabla^2 \phi = 1 \quad (30)$$

- on the mesh provided.
- The boundary conditions are:

$$\phi = 0 \quad (31)$$

- On ALL boundaries (The membrane problem).